# SPE-229644-MS

# Forecast of Stimulation Performance and $CO_2$ Sequestration Capacity of CCUS-EGR Technology for Tight Sandstone Gas Reservoirs in Ordos Basin, NW China

Gang Cheng, Ailin Jia, Zhi Guo, Kefei Chen, Jianlin Guo, Qifeng Wang, Jiangchen Han, Hanqing Zhu, Ruohan Liu, Suqi Huang, and Zhaolong Liu, Research Institute of Petroleum Exploration and Development, PetroChina, Beijing, China

 DOI 10.2118/229644-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.



## Abstract

Under the background of net-zero emissions, CCUS-EGR (carbon capture, utilization and storage & enhanced gas recovery) technology serves as a critical approach for enhancing gas recovery of tight gas reservoirs while reducing greenhouse gas emissions. Currently, the application of CCUS-EGR technology for tight gas reservoirs still remains the early stage of field trails, and it is necessary to evaluate the enhanced gas recovery efficiency and the carbon sequestration capacity of CCUS-EGR. Therefore, through PVT (pressure-volume-temperature) property analysis of $CH_4$ (methane) and $CO_2$ (carbon dioxide), laboratory-scale CCUS-EGR experiments using tight sandstone rock samples, theoretical equation derivation for recovery factor prediction, and numerical simulations of CCUS-EGR in tight gas reservoirs, the stimulation efficiency and $CO_2$ sequestration capacity of CCUS-EGR in tight gas reservoirs in Ordos Basin were systematically analyzed. The following conclusions were drawn: (1) The density and viscosity of supercritical $CO_2$ are significantly greater than those of $CH_4$, and the diffusivity of $CO_2$ under reservoir condition is relatively low, which is advantageous for its efficient displacement of $CH_4$ within the gas reservoir. (2) After depleting rock cores to abandonment pressures of 10.0 MPa, 7.5 MPa, and 5.0 MPa, $CO_2$ was injection into rock cores. When $CO_2$ breakthrough occurred, the resulting recovery factor increments could reach 25.6%, 17.4%, and 10.7%, respectively; (3) The recovery factor increment of CCUS-EGR is mainly affected by four factors including sweep efficiency of pressure drop, depletion efficiency in depletion development stage, $CO_2$ sweep efficiency and $CO_2$ displacement efficiency. Because of the reservoir heterogeneity and gas channeling, these four factors are 47.2%, 61.0%, 38.5% and 87.3% respectively in the X pilot area, resulting in an estimated recovery factor increment of only 4.8%, which is close to the numerical simulation result (5.1%) and lower than the experimental result (25.6%); (4) Based on the proportional conversion of reserves in the pilot area, it is estimated that CCUS-EGR technology will increase tight gas production in Ordos Basin by approximately $1.3\times10^{11}$ $m^3$ and sequestrate about $8.49\times10^8$ tons of $CO_2$. This paper systematically investigates the mechanisms of enhanced gas recovery and storage

capacity of CCUS-EGR in tight gas reservoirs. The findings offer a valuable reference for promoting the application of CCUS-EGR technology in tight gas reservoirs in Ordos Basin, Northwest China.

**Keywords:** Net-zero emissions, Tight gas reservoir, CCUS-EGR, PVT analysis, Physical simulation experiment, Numerical simulation

## Introduction

Under the background of net-zero emissions, CCUS-EGR has attracted increasing attention in the natural gas industry. CCUS-EGR can not only sequestrate $CO_2$ to reduce greenhouse gas emissions, but also significantly improves gas reservoir ultimate recovery. The cumulative proven reserves of tight gas in Ordos Basin, Northwest China are about $5.5\times10^{12}$ $m^3$, and the current average recovery factor in developed areas is only about 32%. After the application of major technical means such as increasing infilling wells, artificially lifting annular produced water, and installing gas compressors to lower wellhead pressure, the recovery factor can be increased by 10% to 15%. However, a large amount of residual gas between wells and in different layers still cannot be produced, so it is urgent to apply CCUS-EGR as an aggressive production stimulation technology to effectively extract the residual gas [1–3]. Currently, CCUS-EGR technology is still in the stage of small-scale field trails all over the world, and countries such as China, the Netherlands, Germany, and Hungary have carried out several field trails and applications [4–8]. The Budafa gas field in Hungary injected $CO_2$ into a depleted gas reservoir with a depletion recovery factor of 67.0%, and it is expected to increase the gas recovery factor by 11.6% through CCUS-EGR. The Wolonghe gas field in the Sichuan Basin, Southwest China is carrying out a CCUS-EGR filed trail in carbonate gas reservoirs, which is expected to increase the recovery factor by about 10% [9–10]. The X pilot area in Ordos Basin is one of the earliest lithologic trap tight sandstone gas reservoirs which implements CCUS-EGR to enhance gas recovery in China. The test formations are the He 8 and Shan 1 members of the Paleozoic formation, with an area of 8.7 $km^2$ and an OGIP of $1.43\times10^9$ $m^3$. The formation temperature is 95.0 ℃, and the initial formation pressure is 27.0 MPa, and current formation pressure is nearly 12.0 MPa. Different from conventional gas reservoirs, the tight sandstone gas reservoirs in Ordos Basin have stronger reservoir heterogeneity, lower porosity (9.3%) and permeability (0.72 mD). Therefore, it is more challenging to carry out the CCUS-EGR in tight sandstone gas reservoirs in the X pilot area.

As a new technology to enhance gas recovery of tight gas reservoirs, CCUS-EGR faces many problems, including unclear mechanism of enhanced gas recovery, lack of an accurate evaluation method of recovery factor increment and $CO_2$ sequestration capacity for CCUS-EGR in tight gas reservoirs [11–12]. To address these issues, this paper conducts PVT property analysis of $CH_4$ and $CO_2$, laboratory-scale experiments, equation derivation for recovery factor increment prediction, and numerical simulation of CCUS-EGR in the X pilot area. Therefore, the stimulation performance and $CO_2$ sequestration capacity of CCUS-EGR in tight gas reservoirs are systematically analyzed, which provides a reference for the large-scale application of CCUS-EGR technology in tight gas reservoirs in Ordos Basin, Northwest China.

## Analysis of PVT Properties of $CH_4$ and $CO_2$

Based on the PVT property analysis of $CH_4$ and $CO_2$, the variations of density, viscosity, Z-factor and diffusivity of $CH_4$ and $CO_2$ under different temperature and pressure were measured (Figure 1 to Figure 4). When the pressure is constant, the density, viscosity and diffusivity of both $CH_4$ and $CO_2$ decrease with the increase of temperature, but the Z-factor of both $CH_4$ and $CO_2$ increases with the growth of temperature. When the temperature is constant, the density and viscosity of both $CH_4$ and $CO_2$ increase with the increase of pressure, while the Z-factor first decreases and then increases with the increase of pressure; and the diffusivity of $CO_2$ decreases with the increase of pressure. Under current reservoir condition (temperature 95.0 ℃, pressure 12.0 MPa), $CO_2$ has reached the supercritical state (critical temperature

31.26 ℃, critical pressure 7.38 MPa). The density and viscosity of supercritical $CO_2$ are significantly greater than those of $CH_4$. The density and viscosity differences between supercritical $CO_2$ and $CH_4$ increase with rising pressure, but decrease with increasing temperature. Under reservoir conditions, the density of supercritical $CO_2$ is about 4.0 times that of $CH_4$. Because the density of supercritical $CO_2$ is much higher than $CH_4$, $CO_2$ tends to migrate downward in the formation. When the formation has a certain dip angle, the gravitational differentiation between $CO_2$ and $CH_4$ is conducive to stable displacement of natural gas. Under reservoir conditions, the viscosity of supercritical $CO_2$ is about 1.8 times that of $CH_4$. Since the viscosity of supercritical $CO_2$ is greater than that of $CH_4$ and its mobility is lower than that of $CH_4$, the injected $CO_2$ does not substantially mix with $CH_4$, thus avoiding obvious miscible displacement and viscosity fingering, which is advantageous for its efficient displacement of $CH_4$ within the reservoir.

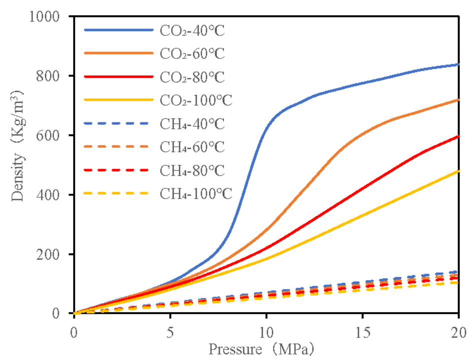



Figure 1—Density of $CH_4$ and $CO_2$.

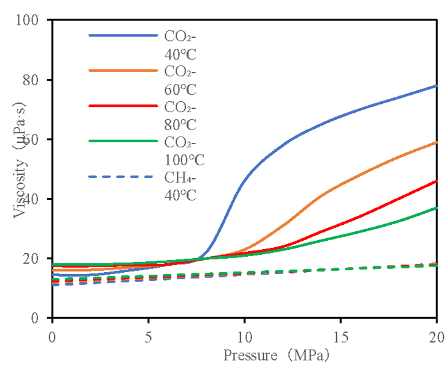



Figure 2—Viscosity of $CH_4$ and $CO_2$.

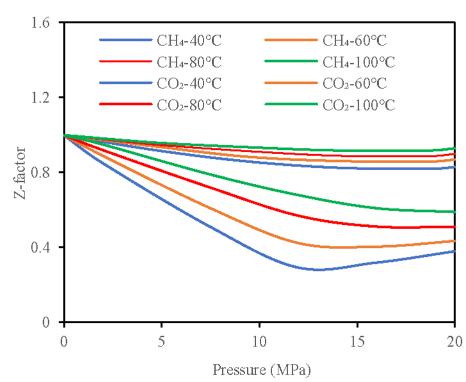



Figure 3—Z-factor of $CH_4$ and $CO_2$.

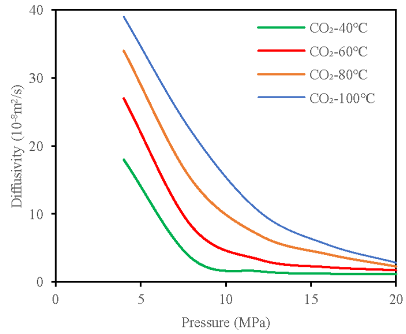



**Figure 4—Diffusivity of $CH_4$ and $CO_2$.**

Under reservoir conditions, the Z-factor of supercritical $CO_2$ is smaller than $CH_4$, indicating that $CO_2$ is more compressible than $CH_4$. Because the Z-factor ratio is equivalent to the volumetric factor ratio of different gases. Therefore, if the formation pressure is maintained at 10.0 MPa to 15.0 MPa, to displace 1 $m^3$ of $CH_4$, it is expected that at least 1.3 $m^3$ to 1.5 $m^3$ of $CO_2$ need to be injected. In addition, the diffusivity of supercritical $CO_2$ under the reservoir condition (approximately $8.2\times10^{-8}$ m/s) is much lower than that under the surface condition (approximately $2.0\times10^{-5}$ m/s), indicating that the $CO_2$ injected into formations does not mix significantly with $CH_4$, which is also conducive to the stable displacement. The special physical properties of supercritical $CO_2$ are beneficial for its stable displacement of $CH_4$ in formations, thereby efficiently improving the gas recovery [13].

## Experiments of CCUS-EGR in Tight Gas Reservoirs

### Experimental Process and Rock Sample Parameters

An experimental device for simulating the CCUS-EGR process was designed to evaluate the stimulation efficiency and carbon sequestration capacity of CCUS-EGR in tight gas reservoirs[14]. The experimental device mainly consists of three parts: a high-pressure gas injection system, a core holding test system, and an injected gas breakthrough monitoring and collection system (Figure 5). The experimental temperature is 95 °C, with an experimental pressure of 27.0 MPa. The experimental rock samples are tight sandstone rocks from the pilot area, with a core diameter of 10.0 cm, length of 20.5 cm, porosity of 10.5%, permeability of 0.25 mD, and initial water saturation of 53.5%. Furthermore, experiments of CCUS-EGR can be divided into two stages: the depletion development stage and the $CO_2$ flooding stage. Firstly, the tight sandstone cores were depleted to 10.0 MPa, 7.5 MPa, and 5.0 MPa in the depletion stage. Subsequently, $CO_2$ was injected to evaluate the stimulation efficiency and $CO_2$ sequestration capacity during the $CO_2$ flooding stage.

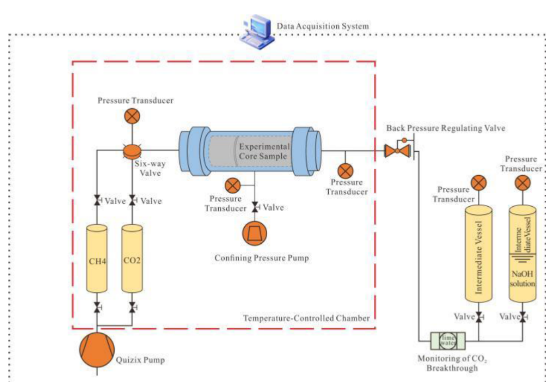
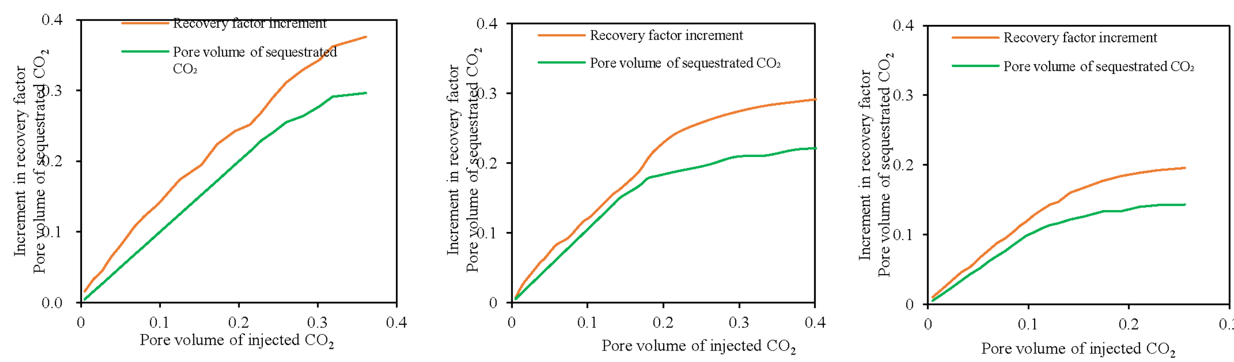



**Figure 5—Experimental device for simulating CCUS-EGR in tight gas reservoirs.**

### Experimental Results and Discussion

Firstly, three rock cores with same petrophysical properties were used to simulate the depletion development of tight gas reservoirs from the initial formation pressure of 27.0 MPa to the abandonment pressures of 10.0 MPa, 7.5 MPa, and 5.0 MPa respectively. At the end of depletion development stage, the $CH_4$ recovery factors were 61.0%, 69.5%, and 78.2% respectively. Then, $CO_2$ was injected into the rock cores to simulate the $CO_2$ flooding process. The $CH_4$ recovery factor increment and sequestrated volume of $CO_2$ increased rapidly with the injection of $CO_2$. When $CO_2$ injection volumes reached 0.22PV, 0.16PV, and 0.09PV, $CO_2$ breakthrough occurred, and the recovery factor increment reached 25.6%, 17.4%, and 10.7%, with $CO_2$ storage volume reaching 0.22PV, 0.16PV, and 0.09PV respectively. After $CO_2$ breakthrough, the $CO_2$ mole fraction in produced gas rose sharply, and the growth rates of recovery factor increment and $CO_2$ storage volume decreased rapidly. When $CO_2$ injection volumes reached 0.36PV, 0.29PV, and 0.18PV, the $CO_2$ mole fraction in produced gas reached 100.0%, with the recovery factor increment reaching 37.9%, 29.6%, and 20.1%, and the $CO_2$ storage volume reaching 0.29PV, 0.22PV, and 0.14PV respectively. According to the experimental results, CCUS-EGR could increase the recovery factor by 10.7% to 37.9%, demonstrating that $CO_2$ was a highly efficient displacement fluid. CCUS-EGR not only enables $CO_2$ sequestration in formations but also significantly enhances gas recovery. Furthermore, lower abandonment pressure will lead to a greater increment in recovery factor (Figure 6 and Table 1).




(a) Abandonment pressure 10.0 MPa (b) Abandonment pressure 7.5 MPa (c) Abandonment pressure 5.0 MPa

**Figure 6—$CH_4$ recovery factor and $CO_2$ storage volume under different abandonment pressures**

**Table 1—Experimental results of CCUS-EGR in tight gas reservoirs**

| Abandonment pressure (Mpa) | Recovery factor in the depletion development stage | Pore volume of injected $CO_2$ at the breakthrough point | Pore volume of injected $CO_2$ while $CO_2$ mole fraction reached 100% | Recovery factor at the breakthrough point | Recovery factor while $CO_2$ mole fraction reached 100% | Pore volume of storage $CO_2$ at the breakthrough point (PV) | Pore volume of storage $CO_2$ while $CO_2$ mole fraction reached 100% (PV) |
|---|---|---|---|---|---|---|---|
| 10 | 61.0% | 0.22 | 0.36 | 25.6% | 37.9% | 0.22 | 0.29 |
| 7.5 | 69.1% | 0.16 | 0.29 | 17.4% | 29.6% | 0.16 | 0.22 |
| 5 | 78.2% | 0.09 | 0.18 | 10.7% | 20.1% | 0.09 | 0.14 |

## Prediction of Recovery Factor Increment of CCUS-EGR in Tight Gas Reservoirs

### Recovery Factor Increment

The experiments of CCUS-EGR with rock samples were based on ideal assumptions such as homogeneous formation, 100.0% formation connectivity, and 100.0% volumetric sweep efficiency of injected $CO_2$. The measured recovery factor increment was the upper limit under ideal conditions. Actually, tight gas reservoirs in Ordos Basin have strong heterogeneity, poor connectivity, and the $CO_2$ injected into formations is prone to cause inter-well gas channeling, resulting in a low sweep efficiency of $CO_2$ in formations. Therefore, to predict the performance of CCUS-EGR on enhancing recovery factor in highly heterogeneous reservoir, a series of equations for predicting the recovery factor increment of CCUS-EGR in tight gas reservoirs were derived:

$$\Delta\eta = E_D E_P (1 - E_V) + E_D E_V E_C - E_D E_P \tag{1}$$

$$E_D = \frac{G_D}{G} \tag{2}$$

$$E_V = \frac{G_V}{G_D} \tag{3}$$

$$E_P = 1 - \frac{P_a Z_i}{P_i Z_a} \tag{4}$$

Where, $\Delta\eta$ is the recovery factor increment during $CO_2$ flooding stage, dimensionless; $E_D$ is the sweep efficiency of pressure drop in the depletion development stage, dimensionless; $E_P$ is the depletion efficiency in the depletion development stage, dimensionless; $E_V$ is the $CO_2$ sweep efficiency, dimensionless; $E_C$ is the $CO_2$ displacement efficiency, dimensionless; $G_D$ is the dynamic reserves of gas wells, $m^3$; $G$ is the original gas in place, $m^3$; $G_V$ is the dynamic reserves of gas wells swept by $CO_2$, $m^3$; $P_a$ is the abandonment pressure in the depletion development stage, MPa; $P_i$ is the initial formation pressure, MPa; $Z_a$ is the Z-factor corresponding to the abandonment pressure, dimensionless; $Z_i$ is the Z-factor corresponding to the initial formation pressure, dimensionless.

Simplifying Equation (1) as:

$$\Delta\eta = E_D E_V (E_C - E_P) \tag{5}$$

Equation (5) demonstrates that the recovery factor increment during $CO_2$ flooding stage is mainly affected by four factors. It is positively correlated with the sweep efficiency of pressure drop, $CO_2$ sweep efficiency, and $CO_2$ displacement efficiency, while negatively correlated with the depletion efficiency in the depletion development stage. The pressure drop sweep efficiency and depletion efficiency in the depletion development stage can be directly calculated through Equations (2) and (4). However, the other two parameters of $CO_2$ sweep efficiency and $CO_2$ displacement efficiency are difficult to determine. In order to figure out the calculation method of these two parameters, more detail will be shown in the following paragraphs.

## Displacement Efficiency of $CO_2$

$CO_2$ displacement efficiency is a new parameter, and previous researches have not involved the calculation method of this parameter. Therefore, this paper innovatively proposed a new method to obtain $CO_2$ displacement efficiency based on experimental results. Under the experimental conditions, since both $E_D$ and $E_V$ in equation (5) are 100.0%, the $CO_2$ displacement efficiency $E_C$ can be calculated by substituting $\Delta\eta$ and $E_P$ shown in Table 1 into Equation (5). For example, when the formation pressure decreases to the abandonment pressure pf 10.0 MPa, 7.5 MPa, and 5.0 MPa in the depletion development stage, the depletion efficiencies of pressure drop ($E_P$) were 61.0%, 69.1%, and 78.2% respectively. Then, $CO_2$ flooding was implemented until $CO_2$ breakthrough occurred, the recovery factor increments of $CO_2$ flooding stage ($\Delta\eta$) were 25.6%, 17.4%, and 10.7% respectively. Furthermore, both $E_D$ and $E_V$ were 100.0% in laboratory-scale experiments. By substituting these four factors into Equation (5), it is easy to obtain that the $CO_2$ displacement efficiencies at the critical point of $CO_2$ breakthrough were 86.6%, 86.5%, and 88.9% respectively, with an average value of 87.3%.

Similarly, the $CO_2$ displacement efficiency at different times can also be calculated. The calculation results show that $CO_2$ displacement efficiency is positively correlated with the mole fraction of $CO_2$ in produced gas. The higher the mole fraction of $CO_2$ in produced gas is, the less residual $CH_4$ in the swept area remains in formations, and the higher the $CO_2$ displacement efficiency is. At the critical point of $CO_2$ breakthrough, the average $CO_2$ displacement efficiency was about 87.3%. When the mole fraction of $CO_2$ in produced gas reached 50%, the average $CO_2$ displacement efficiency was about 95.8%. When the mole fraction of $CO_2$ in produced gas reached 100%, the average $CO_2$ displacement efficiency was about 98.6%. It is concluded that $CO_2$ has a high displacement efficiency and is a relatively efficient displacement fluid.

## Sweep Efficiency of $CO_2$

***Natural Gas Reserves Swept by $CO_2$.*** Tight sandstone gas reservoirs in Ordos Basin have strong reservoir heterogeneity, and the injected $CO_2$ is prone to cause inter-well gas channeling, resulting in great uncertainty in calculating the actual natural gas reserves swept by $CO_2$. Therefore, before calculating the $CO_2$ sweep efficiency through Equation (3), it is necessary to calculate the natural gas reserves swept by $CO_2$ first. As shown in Section 3, after $CO_2$ breakthrough, the $CO_2$ mole fraction in produced gas will rise rapidly, the growth rates of $CH_4$ recovery factor and $CO_2$ storage volume will decrease rapidly. In addition, the $CO_2$ in produced gas after breakthrough will severely corrode pipelines. Therefore, $CO_2$ breakthrough is taken as the critical point to shut-in producers, and the natural gas reserves swept by $CO_2$ reaches its maximum value at this critical point.

In order to accurately predict the $CO_2$ breakthrough time, a new model which takes into consideration the reservoir heterogeneity was proposed. In the new model, it is assumed that the fluid low is linear between a injector and a producer. As shown in Figure 7, $L$ is the distance between the injector and the producer, $\varphi$ is the porosity, $K$ is the permeability, $S_g$ is the gas saturation, and $h$ is the net pay thickness.

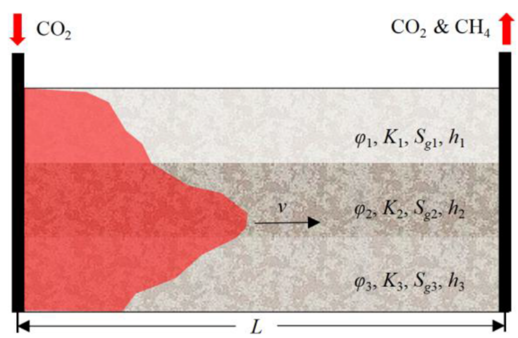



**Figure 7—Schematic diagram of the $CO_2$ swept area between a injector and a producer**

For the high-velocity non-Darcy flow, the pressure drop and gas flow rate can be expressed with the Forchheimer equation:

$$-\frac{dP}{dx} = \frac{\mu}{K}v + \beta\rho v^2 \tag{6}$$

where $P$ is the pressure, Pa; $x$ is the distance, m; $\mu$ is the viscosity of $CO_2$, Pa·s; $K$ is the reservoir permeability, m$^2$; $v$ is the flow rate of $CO_2$, m/s; $\beta$ is the gas turbulence factor of non-Darcy flow, m$^{-1}$; $\rho$ is the density of $CO_2$, kg/m$^3$.

From Equation (6), the average $CO_2$ flow rate between the injector and the producer can be calculated as:

$$v = \frac{-\frac{\mu}{K} + \sqrt{\left(\frac{\mu}{K}\right)^2 - \frac{4\beta\rho\Delta P}{L}}}{2\beta\rho} \tag{7}$$

where $L$ is the distance between the injector and the producer, m; $\Delta P$ is the pressure drop between the injector and the producer, Pa.

Then, the breakthrough time can be determined:

$$t_c = \frac{L}{v} = \frac{2\beta\rho L}{-\frac{\mu}{K} + \sqrt{\left(\frac{\mu}{K}\right)^2 - \frac{4\beta\rho\Delta P}{L}}} \tag{8}$$

Due to the strong heterogeneity of tight gas reservoirs in Ordos Basin, reservoir permeability contrast, half length and flow conductivity of hydraulic fractures will all affect the $CO_2$ breakthrough time. In addition, there is gas-water two-phase flow in formations, so the relative permeability of gas phase will also affect the $CO_2$ breakthrough time. Therefore, considering the influence of all the above factors, a modified equation for calculating pseudo-permeability can be established:

$$K_e = K_{\mathrm{avg}}\left(\frac{K_{\max}}{K_{\min}}\right)^{\alpha}\left(1+\varepsilon\frac{l_f}{L}\log\left(\frac{K_f w_f}{K_{\mathrm{avg}} r_w}\right)\right)K_r \tag{9}$$

where $K_e$ is the pseudo-permeability considering reservoir heterogeneity, m$^2$; $K_{avg}$ is the average matrix permeability, m$^2$; $K_{max}$ is the maximum matrix permeability, m$^2$; $K_{min}$ is the minimum matrix permeability, m$^2$; $l_f$ is the half length of hydraulic fractures, m; $K_f$ is the fracture permeability, m$^2$; $K_r$ is the relative permeability of gas phase, m$^2$; $w_f$ is the width of hydraulic fractures, m$^2$; $r_w$ is the wellbore radius, m; $\alpha$ is the coefficient correlated with permeability contrast, dimensionless; $\varepsilon$ is the coefficient correlated with fracture permeability, m$^2$; Both $\alpha$ and $\varepsilon$ can be obtained by fittings of numerical simulation results and experimental results.

Then the modified $CO_2$ breakthrough time can be expressed:

$$t_{ce} = \frac{L}{24\times 3600\times v} = \frac{2.31\times 10^{-5}\beta\rho L}{-\frac{\mu}{K_e} + \sqrt{\left(\frac{\mu}{K_e}\right)^2 - \frac{4\beta\rho\Delta P}{L}}} \tag{10}$$

where $t_{ce}$ is the $CO_2$ breakthrough time considering reservoir heterogeneity, d.

Assuming that the $CO_2$ injection process is radial flow, the $CO_2$ injection rate satisfies the following equation:

$$q = \frac{-A + \sqrt{A^2 + 4A\left(P_c^2 - P_r^2\right)}}{2B} \tag{11}$$

$$A = 42.42\times 10^3\frac{\bar{\mu}\bar{Z}TP_{sc}}{K_c h T_{sc}}\left(\lg\frac{8.091\times 10^{-3}K_c t}{\varphi\bar{\mu}C_t r_w^2} + 0.8686S\right) \tag{12}$$

$$B = 36.85\times 10^3\frac{\bar{\mu}\bar{Z}TP_{sc}}{K_c h T_{sc}}D \tag{13}$$

$$D = \frac{2.132\times 10^{-3} r_c K_c^{0.066}}{\bar{\mu} h r_w} \tag{14}$$

Where, $q$ is the injection rate, $10^4$ m$^3$/d; $P_c$ is the $CO_2$ injection pressure, MPa; $P_r$ is the formation pressure, MPa; $P_{sc}$ is the pressure under the standard condition, MPa; $\bar{\mu}$ is the average viscosity of $CO_2$ under reservoir conditions, mPa·s; $\bar{Z}$ is the Z-factor under reservoir conditions, dimensionless; $T$ is the reservoir temperature, K; $T_{sc}$ is the temperature under the standard condition, K; $K_c$ is the gas phase permeability, mD; $h$ is the net pay thickness of gas layers, m; $t$ is the duration of $CO_2$ injection, h; $\varphi$ is the porosity of gas layers, dimensionless, $C_t$ is the rock comprehensive elastic compressibility coefficient, MPa$^{-1}$; $S$ is the skin factor, dimensionless; $D$ is the non-Darcy flow coefficient, ($10^4$ m$^3$/d)$^{-1}$; $r_c$ is the relative density of $CO_2$, dimensionless.

As shown in Section 2, the Z-factor factor of $CH_4$ under reservoir condition is about 1.4 times that of $CO_2$. Therefore, if a injection-production ratio of 1.4:1 is adopted during the $CO_2$ injection stage, the formation pressure can be maintained at the current formation pressure and kept balanced. Therefore, the pressure difference between $P_c$ and $P_r$ in Equation (11) can be calculated, thereby the injection rate also being determined. Consequently, the cumulative $CO_2$ injection volume at breakthrough point is:

$$Q = 10000 \times \sum_{1}^{n} \int_{o}^{t_c} q dt \tag{15}$$

Where, $Q$ is the cumulative volume of injected $CO_2$, m$^3$; $n$ is the number of injectors, dimensionless.

In addition, at the critical point of $CO_2$ breakthrough, there are two zones between the injector and the producer: a pure $CO_2$ zone and a $CO_2$ and $CH_4$ mixture zone (Figure 8). Assuming that the gas components in the $CO_2$ and $CH_4$ mixture zone change linearly and the length of mixture zone is $M$, the original $CH_4$ reserves minus the residual $CH_4$ reserves equals the cumulative production of $CH_4$.

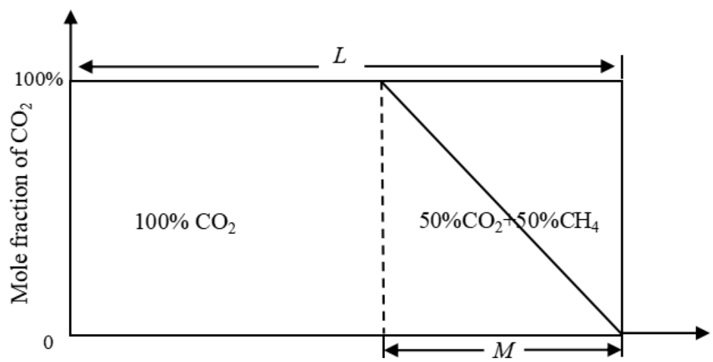



**Figure 8—The mole fraction distribution of $CO_2$ and $CH_4$ between a injector and a producer**

$$\frac{LWh\varphi S_g}{B_{gi}} - \frac{1}{2} \times \frac{MWh\varphi S_g}{B_g} = \frac{LWh\varphi S_g E_C}{B_{gi}} \tag{16}$$

Where $S_g$ is the reservoir gas saturation, dimensionless; $W$ is the width of the area swept by $CO_2$, m; $M$ is the length of $CO_2$ and $CH_4$ mixture zone, m; $B_{gi}$ is the volumetric coefficient of $CH_4$ under the initial reservoir condition, dimensionless; $B_g$ is the volumetric coefficient of $CH_4$ under the current reservoir condition, dimensionless.

Equation (16) can be simplified as:

$$M = \frac{2(1 - E_C)LB_g}{B_{gi}} \tag{17}$$

Substituting the experimental results in Table 1 (at an abandonment pressure of 10.0 MPa, the sweep efficiency of $CO_2$ was 87.3%) into Equation the (17), it is easy to obtain $M \approx 0.24L$.

Assuming that the pore volume swept by $CO_2$ is denoted by "$V$" and the fluid low is linear between a injector and a producer, then the relationship between $V$ and $Q$ can be expressed as:

$$Q = \frac{(L - M)}{L} \times \frac{VS_g}{B_c} + \frac{1}{2} \times \frac{M}{L} \times \frac{VS_g}{B_c} \tag{18}$$

Where $V$ is the pore volume swept by $CO_2$, m$^3$; $B_c$ is the volumetric coefficient of $CO_2$ under the current reservoir condition, dimensionless.

Substituting the Equation (17) into the Equation (18), $V$ can be expressed as:

$$V = \frac{QB_c B_{gi}}{S_g\left(B_{gi} - B_g + B_g E_c\right)} \tag{19}$$

The dynamic reserves of natural gas corresponding to the pore volume swept by $CO_2$ can be calculated:

$$G_V = \frac{VS_g}{B_{gi}} = \frac{QB_c}{B_{gi} - B_g + B_g E_c} \tag{20}$$

By substituting the calculated $G_V$ into Equation (3), the $CO_2$ sweep efficiency can be calculated.

***Fitting of Undetermined Coefficients.*** There are two undetermined coefficients in Equation (9), namely $\alpha$ and $\varepsilon$. To determine the values of these two parameters, a numerical simulation model of CCUS-EGR in tight gas reservoirs was established (Figure 9), with an average porosity of 10%, average matrix permeability of 0.7 mD, average gas saturation of 55%. 20 cases with different reservoir heterogeneity conditions were listed in Table 2, and the $CO_2$ breakthrough times of these cases were obtained through numerical simulation. Then, based on the fitting of the numerical simulation results, the values of $\alpha$ and $\varepsilon$ were determined to be 1.03 and 78.09, respectively.

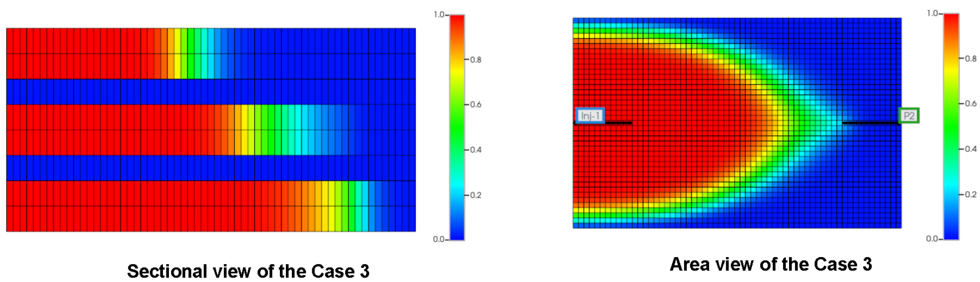



**Figure 9—Gas component distribution during the process of CCUS-EGR (1 represents $CO_2$, 0 represents $CH_4$).**

**Table 2—Numerical simulation results of CCUS-EGR in tight gas reservoirs**

| Numerical simulation | Permeability contrast | Half length of hydraulic fractures (m) | Flow conductivity of hydraulic fractures (D) | Pressure difference between the injector and the producer (d) | Well spacing (m) | $CO_2$ breakthrough time (d) |
|---|---|---|---|---|---|---|
| Case 1 | 1 | 100 | 400 | 7 | 600 | 1796 |
| Case 2 | 2 | 100 | 400 | 7 | 600 | 921 |
| Case 3 | 3 | 100 | 400 | 7 | 600 | 617 |
| Case 4 | 4 | 100 | 400 | 7 | 600 | 463 |
| Case 5 | 2 | 50 | 400 | 7 | 600 | 1079 |
| Case 6 | 2 | 150 | 400 | 7 | 600 | 832 |
| Case 7 | 2 | 200 | 400 | 7 | 600 | 742 |
| Case 8 | 2 | 250 | 400 | 7 | 600 | 597 |
| Case 9 | 2 | 100 | 100 | 7 | 600 | 991 |
| Case 10 | 2 | 100 | 200 | 7 | 600 | 891 |
| Case 11 | 2 | 100 | 300 | 7 | 600 | 886 |
| Case 12 | 2 | 100 | 500 | 7 | 600 | 935 |
| Case 13 | 2 | 100 | 400 | 3 | 600 | 2696 |
| Case 14 | 2 | 100 | 400 | 5 | 600 | 1430 |
| Case 15 | 2 | 100 | 400 | 9 | 600 | 639 |
| Case 16 | 2 | 100 | 400 | 11 | 600 | 472 |
| Case 17 | 2 | 100 | 400 | 7 | 300 | 152 |
| Case 18 | 2 | 100 | 400 | 7 | 400 | 315 |
| Case 19 | 2 | 100 | 400 | 7 | 500 | 582 |
| Case 20 | 2 | 100 | 400 | 7 | 600 | 1247 |

## Numerical Simulations of CCUS-EGR in the Pilot Area

To further elucidate the production stimulation and carbon storage potential of CCUS-EGR in tight gas reservoirs, additional numerical simulations of CCUS-EGR were conducted for the X pilot area [15]. The simulation results were compared with the calculation results of the theoretical equations in Section 4 to validate the accuracy of the theoretical equations.

### Numerical Simulation Model

As of the end of 2024, a total of 22 vertical wells and 2 horizontal wells have been put into production in the pilot area, and the cumulative gas production for these 24 wells were $4.35\times10^8$ $m^3$. In addition, the dynamic reserves of all wells in the pilot area are estimated to be $6.75\times10^8$ $m^3$, with an anticipated ultimate cumulative production of $5.25\times10^8$ $m^3$, projecting a depletion development recovery factor of 36.7%. In the subsequent stage, the gas reservoir will continue to be depleted until the formation pressure decreases to 10.0 MPa. At that time, three producers in the pilot area will be converted into injectors (INJ-1, INJ-2, and INJ-4). Additionally, a new injector (INJ-3) will be drilled in the central region where the well density is relatively low. Then, $CO_2$ will be injected into the formation through these four injectors. The injection pressure for injectors will be maintained at 14 MPa, and the ratio of daily injection volume to daily production volume will be kept at 1.4:1, which ensures that the formation pressure remains stable around 10.0 MPa. Numerical simulations of CCUS-EGR in the X pilot area were conducted by using the CMG software. The numerical simulation model of the gas reservoir is depicted in Figure 10.

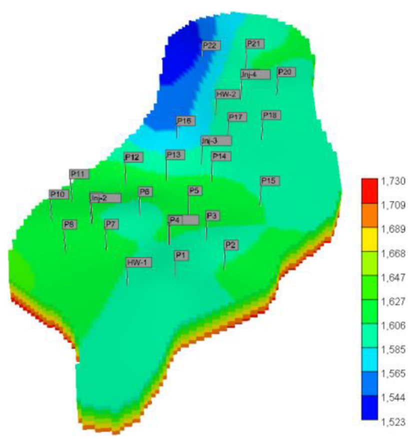



**Figure 10—The numerical simulation model of the X pilot area (True vertical depth sub-sea)**

### Simulation results and analysis

The numerical simulation outcomes indicate that compared with the depletion development, CCUS-EGR will increase the $CH_4$ production of the pilot area by approximately $7.29\times10^7$ $m^3$. The recovery factor increment during $CO_2$ flooding stage is 5.1%, which is lower than the recovery factor increment from experimental results (at an abandonment pressure of 10.0 MPa, the recovery factor increment of CCUS-EGR was 25.6%). This is because tight sandstone gas reservoirs exhibit strong heterogeneity and $CO_2$ is prone to cause inter-well gas channeling, leading to a relatively small $CO_2$ swept area. Based on Equation (2), the sweep efficiency of pressure drop ($E_D$) in the depletion development stage is determined to be 47.2%. According to Equation (19), the dynamic reserves swept by $CO_2$ ($G_V$) is calculated to be $2.60\times10^8$ $m^3$. Then, substituting the dynamic reserves swept by $CO_2$ into Equation (3), $CO_2$ sweep efficiency ($E_V$) is calculated to be 38.5%. As discussed in Section 4.2, the average displacement efficiency of $CO_2$ ($E_C$) at the breakthrough point is approximately 87.3%. In addition, according to Equation (4), the depletion efficiency

($E_P$) in the depletion development stage corresponding to an abandonment pressure of 10.0 MPa is estimated at approximately 61.0%. Finally, based on Equation (5), it is estimated that CCUS-EGR can increase the recovery factor of the pilot area by approximately 4.8%. The recovery factor increment derived from the theoretical equations is close to the numerical simulation results, confirming the accuracy and reliability of the theoretical equations (Figure 11).

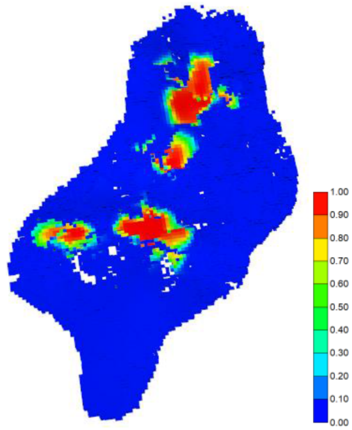



**Figure 11—$CO_2$ distribution during $CO_2$ flooding stage in the X pilot area**

In addition, after the $CO_2$ flooding, it is estimated that a total of 4.76×10$^5$ tons of $CO_2$ will be sequestrated. The $CO_2$ sequestration in the pilot area mainly consists of four parts, with the structural sequestration and mineralization sequestration being the largest, while the residual sequestration and solubility sequestration being relatively small (Figure 12) [16]. It is estimated that after 1000 years, structural sequestration and solubility sequestration will decrease to 2.95×10$^5$ t and 3.96×10$^4$ t, while residual sequestration and mineralization sequestration will rise to 3.56×10$^4$ t, 1.06×10$^5$ t respectively.

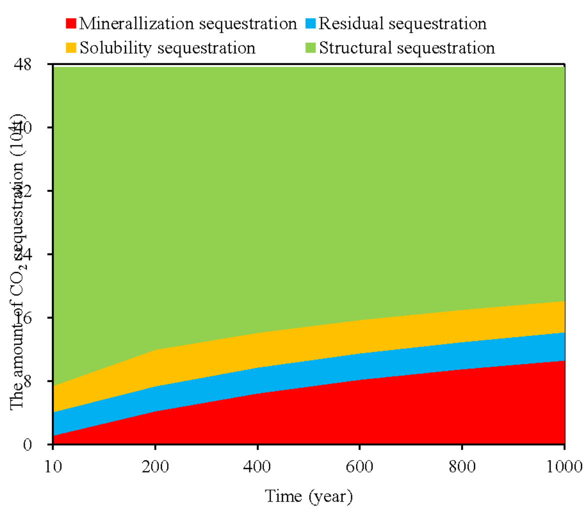



**Figure 12—$CO_2$ sequestration composition in the X pilot area**

Although CCUS-EGR can effectively enhance the recovery factor of gas reservoirs, the recovery factor increment observed in the pilot area (5.1%) has not met the expected target (over 10%) because of the low sweep efficiency of pressure drop (47.2%) and low $CO_2$ sweep efficiency (38.5%). To improve the stimulation performance and $CO_2$ storage of CCUS-EGR in tight sandstone gas reservoirs, future pilot trails should prioritize reservoirs with more injectors, well spacing density and lower reservoir heterogeneity. It is anticipated that under these conditions, the recovery factor increment can exceed 10%. The proven reserves of tight gas reservoirs in Ordos Basin amount to approximately $5.5\times10^{12}$ $m^3$. After excluding areas unsuitable for development, the proven reserves of tight gas reservoirs potentially suitable for CCUS-EGR development are estimated at approximately $1.3\times10^{12}$ $m^3$. Based on the proportional conversion of reserves in the pilot site, it is estimated that CCUS-EGR technology will increase tight gas production in Ordos Basin by approximately $1.3\times10^{11}$ $m^3$ and sequestrate about $8.49\times10^8$ tons of $CO_2$.

## Conclusions

This paper presents a comprehensive study that includes PVT properties analysis, laboratory-scale experiments, theoretical equation derivation and numerical simulation research on CCUS-EGR in tight gas reservoirs. These studies were conducive to ascertain the effectiveness of production stimulation and $CO_2$ sequestration of CCUS-EGR in tight sandstone gas reservoirs in Ordos Basin, Northwest China. The following conclusions have been reached:

1. The density and viscosity of supercritical $CO_2$ are significantly greater than those of $CH_4$, and the diffusivity of $CO_2$ under reservoir condition is relatively low, which indicates that the injected $CO_2$ does not substantially mix with $CH_4$, thus avoiding obvious miscible displacement and viscosity fingering. These unique physical characters of supercritical $CO_2$ are advantageous for its stable displacement of $CH_4$ in formations, thereby efficiently improving the recovery factor of the gas reservoir.
2. In laboratory-scale experiments of CCUS-EGR in tight sandstone reservoirs, after depleting the gas reservoir to abandonment pressures of 10.0 MPa, 7.5 MPa, and 5.0 MPa, $CO_2$ was injected into the gas reservoir. $CO_2$ breakthrough occurred at injection volumes of 0.22 PV, 0.16 PV, and 0.09 PV respectively, resulting in recovery factor increments of 25.6%, 17.4%, and 10.7%. Furthermore, at $CO_2$ injection volumes of 0.36 PV, 0.29 PV, and 0.18 PV, the $CO_2$ mole fraction in produced gas reached 100%, with corresponding recovery factor increments of 37.9%, 29.6%, and 20.1% respectively. The sequestration volumes of $CO_2$ were measured to be 0.29 PV, 0.22 PV, and 0.14 PV respectively.
3. The recovery factor increment is mainly affected by four factors including sweep efficiency of pressure drop, depletion efficiency in depletion development stage, $CO_2$ sweep efficiency and $CO_2$ displacement efficiency. Because of the reservoir heterogeneity and gas channeling, these four factors are 47.2%, 61.0%, 38.5% and 87.3% respectively in the X pilot area, resulting in an estimated recovery factor increment of only 4.8% based on theoretical equations, which is close to the numerical simulation result (5.1%) and lower than the experimental result (25.6%);
4. Based on the proportional conversion of reserves in the pilot site, it is estimated that CCUS-EGR technology will increase tight gas production in Ordos Basin by approximately $1.3\times10^{11}$ $m^3$ and sequestrate about $8.49\times10^8$ tons of $CO_2$.

**Funding**

This research was funded by China National Science and Technology Project (NO. 2025ZD1404300); Project of Science and Technology of PetroChina (NO. 2024DJ86); Project of Science and Technology of PetroChina Oil, Gas and New Energy Company (NO. 2023YQX105).

## References

1. Ma, X., He, D., Wei, Y., et al Enhanced gas recovery: Theory, technology and prospect. *Natural Gas Industry* **43**(1), 1–12 (2023).

2. ZOU Caineng, LIN Minjie, MA Feng, et al Development, challenges and strategies of natural gas industry under carbon neutral target in China[J]. *Petroleum Exploration and Development*, 2024, **51**(2): 418–435.
3. Cao, C., Chen, X., Zhang, L., et al Review of gas reservoir $CO_2$ injection for enhanced recovery and sequestration evaluation methods. *Science Technology and Engineering* **24** (18), 7463–7475 (2024).
4. Behnam T. M., Ashkan J. G., Alv-Arne G. A review of modeling approaches for $CO_2$ injection into depleted gas reservoirs: Coupling transient wellbore and reservoir dynamics. *International Journal of Greenhouse Gas Control* **145**, 104406 (2025).
5. Van Der Meer L, Kreft E & Hartman JJ. K12-B a test site for $CO_2$ storage and enhanced gas recovery. SPE EUROPEC/EAGE Annual Conference, (2005).
6. Cook P. J. The $CO_2$ CRC Otway project in Australia. Melbourne: University of Melbourne, (2013)#
7. Zhang K., Lau H. C., Chen Z. Regional carbon capture and storage opportunities in Alberta, Canada. *Fuel* **322**, 124224 (2022).
8. Zhang L., Xiong W., Zhao Y., et al Mechanism of $CO_2$ injection to enhance gas recovery and carbon storage in depleted bottom-water gas reservoirs. *Natural Gas Industry* **44**(4), 25–38 (2024).
9. Xiong W., Zhao Y., Zhang L. Mechanism study of enhanced gas recovery and $CO_2$ storage in naturally fractured carbonate reservoirs using a dual-permeability model coupled with association equation of state. *Gas Science and Engineering* **139**, 205632(2025).
10. Wen S., Li S., He R., et al Paths and policy suggestions for the construction of CCUS regional center in southwestern China. *Natural Gas Industry* **44**(4), 171–179 (2024).
11. Shi Y., Jia Y., Pan W., et al Potential evaluation on $CO_2$-EGR in tight and low-permeability reservoirs. *Natural Gas Industry* **37**(03), 62–69 (2017).
12. Zhao E., Jin Z., Li Ge., et al Feasibility of $CO_2$ storage and enhanced gas recovery in depleted tight sandstone gas reservoirs within multi-stage fracturing horizontal wells. *Petroleum Science* **21**(6), 4189–4203 (2024).
13. Hou D., Gong F., Chen B., et al Gas recovery enhancement and $CO_2$ storage effects by $CO_2$ flooding in bottom-water sandstone gas reservoir. *Natural Gas Industry* **44**(4), 93–103 (2024).
14. Gao S., Ye L., Liu H., et al An experimental study on recovery factor of watered gas reservoirs by $CO_2$ EOR in the Sebei Gas Field of the Qaidam Basin. *Natural Gas Industry* **43**(12), 55–62 (2023).
15. Zhao E., Jin Z., Li G. Numerical simulation of $CO_2$ storage with enhanced gas recovery in depleted tight sandstone gas reservoirs. *Fuel* **371**(A), 131948 (2024).
16. Yang S., Cai M., Zhang K., et al Research progress and prospect of $CO_2$-water-rock interaction on petrophysical properties of $CO_2$ geological sequestration. *Petroleum Geology and Recovery Efficiency* **30**(6), 80–91 (2023).